Frequency-Calibrated Sinusoidal Positional Encoding

The standard sinusoidal positional encoding (PE) introduced in the Transformer uses sinusoidal functions with geometrically spaced frequencies. It is defined as

$$PE(pos, 2i) = \sin\left(pos \cdot B^{-2i/d_{\text{model}}}\right) \tag{1}$$

$$PE(pos, 2i+1) = \cos\left(pos \cdot B^{-2i/d_{\text{model}}}\right) \tag{2}$$

where pos denotes the token position, i is the frequency index, d_{model} is the embedding dimension, and B is the base parameter (set to 10000 in the original Transformer).

The term $B^{-2i/d_{\text{model}}}$ determines the angular frequency of the sinusoid. The positional encoding can therefore be interpreted as a standard sinusoidal signal

$$\sin(pos \cdot 2\pi f_i) \tag{3}$$

where the frequency associated with dimension i is

$$2\pi f_i = B^{-2i/d_{\text{model}}}. \tag{4}$$

This shows that sinusoidal positional encoding constructs a geometric ladder of frequencies across the embedding dimensions.

Frequency Calibration To better match the temporal structure of the dataset, the base parameter can be calibrated using the dominant frequency of the time series. Let f_{dom} denote the dominant dataset frequency obtained from FFT analysis. If we align this frequency with ladder index k, we obtain

$$2\pi f_{\text{dom}} = B^{-2k/d_{\text{model}}}. \tag{5}$$

Solving for the base yields

$$B = (2\pi f_{\text{dom}})^{-d_{\text{model}}/(2k)}. \tag{6}$$

This produces a family of candidate bases corresponding to different ladder indices k. Each candidate shifts the spectrum of the positional encoding so that one sinusoidal component matches the dominant frequency observed in the data.

Experimental Results We evaluated several calibrated bases derived from the dominant frequency of the ETTh1 dataset ($f_{\text{dom}} = 0.04168$). The resulting forecasting performance measured using WAPE is shown in Table 1.

Base Value	WAPE
14.5809	34.0353
45200.1	34.1665
212.603	34.1947
4.6240	34.2349
Default (10000)	34.3477

Table 1: Performance of sinusoidal positional encoding with calibrated base values.

Discussion The results show that calibrating the base parameter based on the dataset's dominant frequency can improve forecasting performance compared to the default setting. The best performance was obtained with a base of 14.58, which corresponds to aligning the dominant frequency with a mid-range frequency index of the encoding ladder. This suggests that the mid-range positional frequencies are most relevant for capturing the periodic structure present in the ETTh1 dataset.

Overall, these findings indicate that adapting the positional encoding frequency spectrum to the data can provide modest but consistent improvements over the fixed base used in the original Transformer formulation.